costing_CONFIDENCE

Confidence ratings (ordering and works-as-intended)

Two columns for every project. They answer different questions:

- **Order confidence:** how sure we are that you can buy exactly these parts and files and get what the BOM says. **High**: every LCSC part live-checked (identity, package, stock above 500, not end-of-life) and the Gerber/BOM/CPL files pass all automatic gates. **Medium-High**: same, but one or more parts have limited stock (still enough for a prototype run) or a substituted part whose pad geometry is unproven. **Medium**: parts depend on a vendor module, a hand-sourced item or a manual cross-check. **Low**: no board or no verified parts yet.
- **Works-as-intended confidence:** how sure we are that the design does its job when built. An engineering judgement, **not a measurement**: nothing has been built. It weighs (a) how standard the circuit is (datasheet reference design versus novel), (b) the evidence we have (host-run firmware tests, simulation, pin-map checks, datasheet equations), (c) risks that only hardware can reveal (analog front ends, RF, power switching, mains, calibration, uncompiled firmware). **High** would need a built and tested unit, so no project is rated High. **Medium-High**: standard circuit, strong test evidence, low-risk failure modes. **Medium**: expect to find one or two issues in bring-up. **Low-Medium**: significant parts of the behaviour are assumed. **Low**: concept or software only.

Plan for one respin. Even a clean design review usually leaves one or two bring-up findings on a first board; order the small quantity first (5 boards) and keep the files in this repository as the single source of truth.

Project	Order confidence	Works-as-intended	Why (works)	First thing to do
Seeti Counter	**Medium-High** (limited stock on: BZ1 Buzzer 3V (stock 140))	**Medium**	Mic front end and clap/whistle detector are analog and tuned on synthetic audio; the firmware itself passes 115 host checks; the documented limit is that quiet whistles are missed.	Bench-test the microphone gain with real claps before ordering more than 5 boards.
Gas Taraju (LPG cylinder scale)	**Medium-High** (limited stock on: BZ1 Buzzer (stock 140))	**Medium**	HX711 load-cell chain is a standard circuit; accuracy depends on the cell, mounting and calibration, which only a build can show.	Calibrate with a known weight; check drift over a day.
Fridge Door Guard	**Medium-High** (limited stock on: BZ1 Buzzer (stock 140); D4 TVS 5V bidir (stock 175))	**Medium-High**	Reed switch, timer and buzzer: simple circuit, simple firmware; TVS and input conditioning follow standard practice.	Test the reed gap and the 2-minute alarm on the bench.
Bijli Meter Reader	**High** (all parts live-checked, none flagged)	**Medium**	Optical pulse reading of a meter LED depends on the sensor alignment and ambient light; circuit is standard.	Try on a real meter with a shroud before enclosure work.
Bijli Log	**High** (all parts live-checked, none flagged)	**Medium**	Current/voltage logging front end is analog and uncalibrated; mains-adjacent measurement needs isolation care.	Do not connect to mains until a qualified electrician reviews the isolation; start with a bench supply.

Project	Order confidence	Works-as-intended	Why (works)	First thing to do
Chulha Timers	**Medium-High** (limited stock on: BZ1 Buzzer (stock 140))	**Medium-High**	Multi-timer with buttons, LEDs and buzzer; simple circuit; firmware tested on the host.	Check button debounce and the buzzer volume on the first board.
Rain Alert Bell + plate	**Medium-High** (limited stock on: BZ1 Buzzer 85dB (stock 140); D3 TVS 5V bidir (stock 175); D4 TVS 5V bidir (stock 175))	**Medium**	Conductive rain plate corrodes and drifts; electronics are simple and protected with TVS.	Use the plate for a week outdoors before deciding on a coating.
Tap Litre Meter	**Medium-High** (limited stock on: BZ1 Buzzer (stock 140); D4 TVS 5V bidir (stock 175))	**Medium**	Hall/flow-pulse counting is standard; the litres-per-pulse constant depends on the flow sensor.	Calibrate against a measured bucket.
Masala Rack (laser-cut)	**Medium-High** (no electronic parts to source: local laser-cutting or printing)	**Medium-High**	Laser-cut rack: fit depends on kerf; stability limit stated.	Cut a kerf coupon first.
Laptop Stand (laser-cut)	**Medium-High** (no electronic parts to source: local laser-cutting or printing)	**Medium-High**	Laser-cut stand: fit depends on kerf; stability limit stated.	Cut a kerf coupon and load-test.
1 Desi Games Studio (print)	**Medium-High** (no electronic parts to source: local laser-cutting or printing)	**Medium**	Printed game; needs language and fact review before print.	Proofread with native speakers.
2 Kalakriti Automata (laser)	**Medium-High** (no electronic parts to source: local laser-cutting or printing)	**Medium**	Laser-cut automata; frame layouts are provisional.	Cut one frame and test the motion.
3 Prayog STEM lab kit	**High** (all parts live-checked, none flagged)	**Medium-High**	Educational kit with a straightforward MCU board; 40 host checks on the statistics and command code.	Run the kit's own experiments on the first unit.
4 Jal-Rakshak tank controller	**Low** (no board and no verified part list yet)	**Low**	No board; the mains version cannot be made; only a low-voltage prototype is sensible.	Do not build a mains version.
5 Kheti-Mitra irrigation	**Medium-High** (limited stock on: J1 12 V DC in (stock 321))	**Medium**	Irrigation controller with 12 V switching and soil sensing; 22 host checks pass; field conditions (water, sun, surge) are the risk.	Run with a lamp as the load for 48 hours before any valve or pump.
6 Gas-Suraksha leak alarm	**Low** (no board and no verified part list yet)	**Low**	Not certified, safety-critical, no board; firmware logic passes 34 checks.	Do not sell or rely on it for safety.
01 Reflex Raja	**Medium-High** (limited stock on: BZ1 Buzzer (stock 140))	**Medium-High**	Shares the Desi Arcade board; game logic tested with about 80,000 host checks.	Play-test the buttons and the buzzer.
02 Taal Memory	**Medium-High** (limited stock on: BZ1 Buzzer (stock 140))	**Medium-High**	Same board and test base as Reflex Raja.	Play-test the buttons and the buzzer.

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03 Aloo Bomb	**Medium-High** (limited stock on: BZ1 Buzzer (stock 140))	**Medium-High**	Same board; game logic host-tested; the potato-clock mechanics are the unknown.	Test timing with a real potato.
04 Beat Runner	**Medium-High** (limited stock on: BZ1 Buzzer (stock 140))	**Medium-High**	Rhythm game on its own board; logic host-tested; LEDs and buzzer circuits are standard.	Check the LED brightness and button spacing.
05 Rangoli Spiro Gears	**Medium-High** (no electronic parts to source: local laser-cutting or printing)	**Medium-High**	Laser-cut gears; kerf test first.	Cut a kerf coupon.
06 Paper Rocket Launcher	**Medium-High** (no electronic parts to source: local laser-cutting or printing)	**Medium**	Projectile toy; needs lab review; drawings missing.	Get a safety review before any sale.
07 Tilt Maze	**Medium-High** (no electronic parts to source: local laser-cutting or printing)	**Medium**	Gimbal frame drawing missing.	Draw and test the frame.
08 Rubber-band Racer	**Medium-High** (no electronic parts to source: local laser-cutting or printing)	**Medium**	Provisional cut file.	Prototype one.
09 Marble Cricket	**Medium-High** (no electronic parts to source: local laser-cutting or printing)	**Medium**	Provisional cut file; needs calibration drop.	Prototype one.
10 Katha Cubes	**Medium-High** (no electronic parts to source: local laser-cutting or printing)	**Medium**	Illustrations and proofreading are missing.	Finish the artwork and text.
KINETIC motion controller + vibration puck	**High** (all parts live-checked, none flagged)	**Medium**	ESP32-S3 controller with three StepStick sockets and a 24 V to 5 V buck. Motion library (370 checks) and protocol are host-tested; the ESP32 firmware has never been compiled; drivers and motors are bought in. The vibration model is trained on synthetic data only.	Power with a current-limited supply and no drivers; then one driver and one motor at low current.
MITTI-NETRA soil + structure node	**High** (all parts live-checked, none flagged)	**Medium**	Charger, LDO, modem switch, ADXL345, HX711 and soil input on one board; firmware core (23 checks) and the pin map match; the cellular modem is a plug-in breakout whose pinout you must match; the ESP32 skeleton has never been compiled.	Bring up power and charging first; add the modem last and check its pinout against J6.
BioTray vertical-farm controller	**Medium-High** (limited stock on: J2 12 V DC in (stock 321))	**Medium**	12 V low-side drivers, ESP32 and sensors; 22 host checks; calibration constants are placeholders; wet environment.	Dummy loads on all outputs first; seal the enclosure before any water.
BioWatt MFC harvester	**Medium-High** (all parts verified; the BQ25570 stock is limited (about 400) and the hold	**Low-Medium**	BQ25570 with megohm setting dividers computed from datasheet equations; cold start needs at least 600 mV (marginal for one microbial fuel cell); the hold capacitor is X7R where the datasheet asks for low leakage;	Test with a bench supply at 0.8 V first, clean flux thoroughly, then try the fuel cell.

Project	Order confidence	Works-as-intended	Why (works)	First thing to do
	capacitor is a substitute grade)		leakage on a dirty board shifts the thresholds.	
GEONODE foundation monitoring	**Medium** (uses other boards plus bought-in modules; each module's pinout must be checked against its datasheet)	**Low-Medium**	Detector logic and feature code are host-tested (54 checks) but on synthetic vibration; the crack signature is an assumption; node hardware is the MITTI-NETRA board with a LoRa module.	Record real vibration for a week and re-derive the thresholds before trusting any alarm.
AQUAPONICS controller	**Medium** (uses other boards plus bought-in modules; each module's pinout must be checked against its datasheet)	**Low-Medium**	Control and forecasting core passes 13 checks; biological limits are placeholders; no ammonia or nitrite sensing; live animals depend on it.	Run a week with water only; tune limits with an aquaculture expert before any fish.
SORTER bio-robotic platform	**Medium** (uses other boards plus bought-in modules; each module's pinout must be checked against its datasheet)	**Low**	Tracking core passes 28 checks; the vision is a colour-rule baseline tested on synthetic discs; the gripper geometry is untested; machine guarding is your responsibility.	Image real produce under fixed light and replace the classifier before building the conveyor.
AGROVER solar rover	**Medium** (uses other boards plus bought-in modules; each module's pinout must be checked against its datasheet)	**Low-Medium**	Navigation and power core pass 19 simulated-field checks; dead-reckoning only; chassis plates untested with real motors; needs a safety review and an emergency stop.	Drive on blocks first, then slowly in a clear area with the stop in hand.
Jal-Netra (engineering marvel)	**Medium-High** (limited stock on: BZ1 Buzzer (stock 140))	**Low-Medium**	Clamp-on ultrasonic transit-time measurement has a hard analog front end; the manual itself says Gate 1 acoustic test comes first and it is not sellable.	Run the Gate 1 acoustic test with the evaluation setup before any more boards.
ASTRADAND gesture wand (sci-fi project)	**Medium** (substitute addressable LED whose pad geometry is unproven; other parts verified)	**Medium**	IMU gesture recognition validated against a Python reference (46/46) and on synthetic gestures; hardware (haptics, six addressable LEDs) is new; the LED part was replaced with a pin-compatible one whose pads are unproven.	Order 2 boards, check the LED footprint fit, then run the gesture self-test.
Gravity toys (5)	**Medium-High** (no electronic parts to source: local laser-cutting or printing)	**Medium**	Mechanical toys: designs are physical and need prototypes; drawings are partial.	Print or cut one of each and test before batches.
Wave 1 electronics toys	**Low** (no board and no verified part list yet)	**Low**	Firmware passes 76 host checks but there is no board and the MCU is an abstract placeholder.	Wait for the audio platform board (task AP1) before ordering anything.
Prototype packs (10)	**Medium** (breadboard parts from local vendors)	**Medium**	Bench prototype packs on breadboards with 26 host checks; no PCB.	Use as learning kits; do not sell.
DJ / party system	**Low** (no board and no verified part list yet)	**Low**	Plan only; not manufacturable.	Treat as a concept.